

BOM

Part No.	Part name	Qty.	Material/Stock	Specification	Files	Notes
P-01	Body	1	Printed	0.4mm nozzle	STEP/STL/PDF	Remove the built-in support structures and clean up as needed.
C-01	Light Seal Felt Strip	1	Felt	5 mm wide × 254 mm		Adhesive-backed felt recommended. Approx.1mm thick.
C-02	Light Seal Felt Sheet	1	Felt		SVG/PDF	Cut to match template. Adhesive-backed felt recommended. Approx.1mm thick.
M-01	Retaining Clip	1	Steel Round Bar	Ø1	PDF	Bend as shown
M-02	Film Claw	1	Stainless Steel Spring Wire	Ø0.4	PDF	Bend as shown
P-02	Film Claw Arm	1	Printed	0.2mm nozzle	STEP/STL/PDF	Remove the built-in support structures and clean up as needed.
M-03	Film Claw Arm Spring	1	Stainless Steel Spring Wire	Ø0.4	SVG	Bend as shown
P-03	Lever	1	Printed	0.4mm nozzle	STEP/STL	
M-04	Lever Pivot Pin	1	Steel Round Bar	Ø0.8 × 4 mm		Cut to length
P-04	Knob Gear	1	Printed	0.2mm nozzle	STEP/STL	
P-05	Idler Gear	1	Printed	0.2mm nozzle	STEP/STL	
P-06	Front Roller Gear	1	Printed	0.2mm nozzle	STEP/STL	
P-07	Rear Roller Gear	1	Printed	0.2mm nozzle	STEP/STL	
M-05	Roller Gear Pin	2	Steel Round Bar	Ø0.8 × 4.4 mm		Cut to length
M-06	Roller Shaft	2	Steel Round Bar	Ø3	PDF	Cut to length and drill holes
M-07	Roller	2	Aluminium Pipe	OD 7 mm × ID 3 mm	PDF	Cut to length and drill a hole
M-08	Roller Shaft Pin	2	Steel Round Bar	Ø0.8 × 5 mm		Cut to length
P-08	Left Roller Retainer	1	Printed	0.4mm nozzle	STEP/STL	
P-09	Right Roller Retainer	1	Printed	0.4mm nozzle	STEP/STL	
P-10	Shaft Spacer	1	Printed	0.4mm nozzle	STEP/STL	
M-09	Roller Pressure Spring	2	Stainless Steel Spring Wire	Ø0.8	SVG	A binder clip spring is a good substitute.
P-11	Roller Pressure Spring Seat	1	Printed	0.4mm nozzle	STEP/STL	
P-12	Roller Shaft Collar	2	Printed	0.4mm nozzle	STEP/STL	
M-10	Bearing Plate	2	Brass Plate	t0.5	PDF	This part can be made with a drill and files, but a milling machine makes the job much easier.
M-11	Back Panel	1	Metal Sheet	t0.3	STEP/DXF	Cut and bend as shown. Any sheet metal from approximately 0.2 mm to 0.8 mm thick should work.
C-03	Film Pack Spring	2	Steel	Ø9 × 15 mm Ø0.5		Maximum recommended size: Ø9 mm × 16 mm, wire Ø0.6 mm. A free length of 14-16 mm works well.
M-12	Latch Stud	1	Steel Round Bar	Ø2 × 2 mm		Cut to length and weld
C-04	Foam Tape	1	Foam Tape	3 mm thick		
C-05	Slotted Screw	2	Metal	M3 × 12 mm		Any M3 screw will work. Slotted screws are used only for appearance.
C-06	Back Panel Leather	1	Leather		SVG/PDF	Cut to match template

Keep the dimensional accuracy of printed parts within approximately ± 0.1 mm. Recommended print settings: Use a 0.16 mm layer height with 0.4 mm nozzle parts, and 0.10 mm with 0.2 mm nozzle parts.

Use any filament that provides sufficient dimensional accuracy and strength. The prototype was printed using PLA+, but PLA-CF, ABS, nylon, or other suitable materials should also work.

The design is intended as a starting point, not a strict specification. If a different manufacturing method or material works better for your tools and skills, use it. Feel free to use any material you can easily obtain and work with. For parts with sliding contact, choose a material with adequate wear resistance.

Prioritize reliable operation over reproducing the prototype exactly. Minor dimensional or material changes are perfectly acceptable if they improve function or manufacturability.

If you find a simpler, stronger, or easier-to-build solution, don't hesitate to adapt the design. Improvements and alternative implementations are always welcome.